

Loss Minimization of Dual Active Bridge Converter through Design Optimization in CC-CV Mode for Electric Vehicle Battery Charging Applications

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Abstract—The dual active bridge (DAB) converter is the most widely adopted topology for the dc-dc power conversion stage in electric vehicle battery charging owing to its inherent capabilities such as soft switching and galvanic isolation. However, the performance of DAB is compromised when it's operating under a constant-current constant-voltage (CC-CV) charging process. Therefore, the optimal design and operation of the DAB converter become of paramount importance to achieve its efficient operation during CC-CV charging. The existing design methods optimize the converter, particularly in CC mode, and take no notice of the CV mode in the design optimization process. Furthermore, the DAB converter's zero voltage switching (ZVS) operation range is limited during the light-load conditions in CV mode. This article overcomes the aforementioned concerns by proposing a loss minimization approach for the DAB converter through design optimization throughout CC-CV mode. Also, a smooth mode transition from CC to CV is achieved for the DAB converter through the control mechanism. The proposed method is formulated analytically and validated with simulation results.

Index Terms—Battery chargers, CC-CV charging, dc-dc power converters, dual active bridge (DAB) converters, electric vehicles, zero voltage switching

I. INTRODUCTION

The advent of electric vehicle (EV) technology has encouraged research in the domain of efficient EV charging infrastructure. Typically, an EV charger consists of an active front-end converter (AFEC) followed by a dc-dc converter [1]. For safety reasons, the dc-dc converter in EV chargers is preferred to be galvanically isolated from the grid [2]. The dual active bridge (DAB) converter and resonant converters have become the popular choices for the isolated bidirectional dc-dc conversion (IBDC) stage of EV chargers as it offers high power density and light weight [3], [4]. Among them, the DAB converter is widely adopted for the IBDC stages due to its simplicity in control and inherent capabilities such as galvanic isolation and soft switching capability [4]. The DAB converter must be optimally designed such that the conduction and switching losses are minimized throughout the charging process. As the battery charges, the voltage conversion gain (VCG) of the DAB converter varies. The VCG and loading conditions affect the zero voltage switching (ZVS) capability of the DAB converter [5]. To minimize the switching turn-on power losses throughout the charging process, the ZVS operation of the DAB converter must be ensured. ZVS of

the DAB converter can be achieved over the wide operating voltage range by maintaining unity VCG [5].

It is well known that core losses in the inductor and the transformer of the DAB converter amount to a small fraction of the total power losses [6]. The power efficiency of the DAB converter is mainly decided by the conduction power losses of the semiconductor devices as ensuring ZVS operation can result in lower switching power losses [7]. A careful study of the DAB converter reveals that the rms value of the current flowing through the semiconductor switches contributes to conduction power loss, and this in turn depends on the value of leakage inductance and the turns ratio of the high-frequency transformer (HFT) [8]. In the battery charging applications, the nominal value of battery pack voltage decides the turns ratio of the HFT whereas the maximum power rating of the DAB converter decides the value of the leakage inductance of the HFT [5].

Designing the DAB converter using the aforementioned conventional design technique does not result in optimal converter performance over the entire charging process. In this regard, a single objective optimization-based design method using single phase shift (SPS) modulation was reported in [9], which does not incorporate all the charging modes while optimizing the design and performance of the converter. Designing the converter around a single charging mode compromises the performance of the DAB converter in the other modes during charging. Constant-current constant-voltage (CC-CV) charging is the typical battery charging technique adopted for Li-ion batteries in EVs [10]. During the transition from CC to CV charging mode, undesired transients may occur if the transition takes place abruptly. To minimize the voltage/current stress in the DAB converter, a smooth transition between the charging modes must be ensured. Smooth transition methods reported in [11], [12] make use of auxiliary circuits which incur additional component count and associated power losses.

To overcome the aforementioned shortcomings, this paper proposes a loss minimization method for the DAB converter through design optimization in all charging modes. Furthermore, a smooth transition method for the CC-CV charging process is achieved through a control mechanism derived from the small-signal model of the DAB converter. Section II of this paper presents an optimization-based design process of

the DAB converter for CC-CV charging. Section III discusses the seamless mode transition method for switching from CC to CV mode proposed for the DAB converter. Finally, results and discussion is presented in Section IV of this paper followed by the conclusion in Section V.

II. OPTIMAL DESIGN OF THE DAB CONVERTER

A. Performance characterization of the DAB converter

A typical power circuit of the DAB converter is shown in Fig. 1. In Fig. 1, semiconductor switches $S_1 - S_4$ and $S_5 - S_8$ constitute the input and output bridges galvanically isolated by the HFT. The turns ratio and leakage inductance of the HFT is represented by n and L_k , respectively. The input and output voltages of the DAB converter are represented by V_{in} and V_o , respectively.

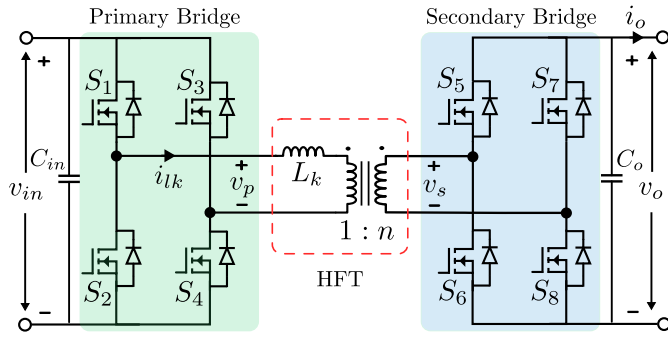


Fig. 1: Power circuit of the DAB converter.

The SPS modulation technique is the most widely used control method for the DAB converter due to its simplicity in both control and soft-switching realization [13]. Typical waveforms showing v_p , v'_s and i_{lk} (Fig. 1) for the DAB converter operated with the SPS modulation method are shown in Fig. 2. From Fig. 2, the expressions for I_1 and I_2 are derived

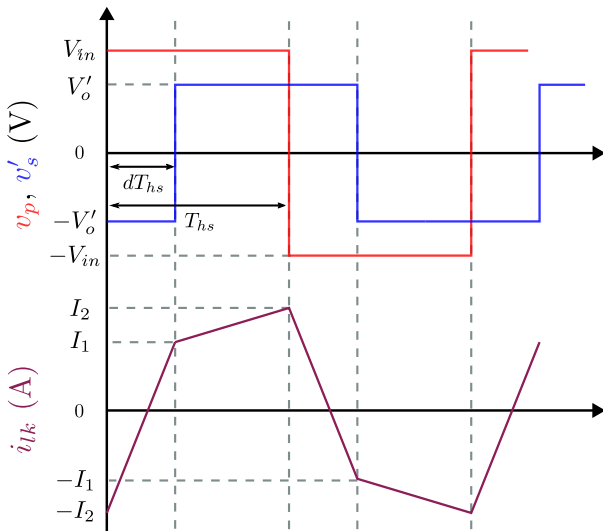


Fig. 2: Operating waveforms of the DAB converter under SPS modulation.

in terms of half-switching time period (T_{hs}) and phase shift ratio (d) and are given in (1) and (2).

$$I_1 = \frac{T_{hs} V_{in}}{2L_k} [(2d - 1) + M] \quad (1)$$

$$I_2 = \frac{T_{hs} V_{in}}{2L_k} [(2d - 1)M + 1] \quad (2)$$

$$M = \frac{V_o}{nV_{in}} \quad (3)$$

Here, M is defined as VCG of the DAB converter.

Using (1)-(3), rms value of i_{lk} i.e. I_{lk} , average values of output current (i_o) and output power (P_o) of the DAB converter are derived and are given in (4)-(6).

$$I_{lk(rms)} = \sqrt{\frac{I_1^2 + I_2^2 + I_1 I_2 (1 - 2d)}{3}} \quad (4)$$

$$I_o = \frac{V_{in} d (1 - d) T_{hs}}{n L_k} \quad (5)$$

$$P_o = \frac{V_o V_{in} d (1 - d) T_{hs}}{n L_k} \quad (6)$$

The ZVS realization of the DAB converter depends on the interaction of parasitic capacitance of the MOSFET and the L_k during the dead-time period when bridges are switched. The condition for realizing ZVS in the X^{th} bridge of the DAB converter is derived and is expressed as [13]

$$I_{lk(BX)} > 2V_{BX} \sqrt{\frac{C_{eq}}{L_k}} \quad (7)$$

In (7), $I_{lk(BX)}$, V_{BX} and C_{eq} are the values of current through L_k , bridge voltage, and equivalent parasitic capacitance of the MOSFET switches, respectively when the bridge is switched.

B. Battery pack specifications

The typical battery capacity of electric two-wheeler (E-2W) is 1.2-3.3 kWh [14]. The DAB converter is intended to charge an E-2W of 3 kWh capacity. In CC mode, the battery is charged with constant current until its terminal voltage reaches a maximum permissible value $V_{o(max)}$ and is then followed by CV mode where the battery voltage is maintained at $V_{o(max)}$ and the current is allowed to decrease below the cut-off value. The battery pack is realized using Panasonic's 18650PF lithium-ion cell [15] whose CC-CV charging curve is shown in Fig. 3. CREE's C3M0065100K Silicon Carbide MOSFET devices are used in this work [16]. The detailed specifications of the DAB converter and the battery pack are tabulated and presented in Table I.

C. Formulation of optimization problem

The conduction power losses in the DAB converter are proportional to the square of I_{lk} [8]. The total conduction

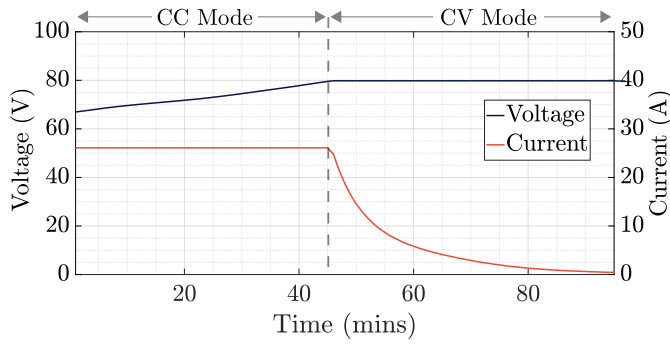


Fig. 3: CC-CV charging profile of the battery pack.

power losses in the DAB converter are derived and is given in (8).

$$P_{cond} = 4R_{on(pri)} \left(\frac{I_{lk}}{\sqrt{2}} \right)^2 + \frac{4R_{on(sec)}}{n^2} \left(\frac{I_{lk}}{\sqrt{2}} \right)^2 \propto I_{lk}^2 \quad (8)$$

Here, $R_{on(pri)}$ and $R_{on(sec)}$ denote the on-state drain-source resistance of primary and secondary bridges, respectively. The expression for total switching power losses in the DAB converter is given in (9).

$$P_{swit} = 4f_s(E_{on(p)} + E_{on(s)}) + 4f_s(E_{off(p)} + E_{off(s)}) \quad (9)$$

Where, $E_{on(p)}$ and $E_{off(p)}$ represent the turn-on and turn-off switching energies of primary bridge MOSFET devices. Similarly, $E_{on(s)}$ and $E_{off(s)}$ represent the turn-on and turn-off switching energies of secondary bridge MOSFET devices. From the detailed discussions in this paper, the total power losses in the DAB converter system mainly depend on I_{lk} ; hence the value of I_{lk} must be optimized while realizing ZVS operation during the entire charging process [6].

The optimization process proposed in this paper is bifurcated into two parts. Firstly, the DAB converter is designed for CC mode to get the optimal values of n and L_k which corresponds to the minimum average rms inductor current

TABLE I: DAB Converter and Battery Pack Specifications

Parameter	Value
Nominal cell voltage $V_{c(nom)}$	3.7 V
Maximum cell voltage $V_{c(max)}$	4.2 V
Nominal battery voltage $V_{o(nom)}$	70.3 V
Maximum battery voltage $V_{o(max)}$	79.8 V
Battery Energy Capacity E_b	3 kWh
CC mode current $I_{o(cc)}$	25.5 A
Maximum Charging Power $P_{o(max)}$	2 kW
CC mode input dc voltage $V_{in(CC)}$	400 V
Switching frequency f_s	50 kHz

($\bar{I}_{lk}$) throughout the CC mode. Secondly, to optimize the performance in CV mode, the input dc-link voltage reference of the DAB converter is changed such that VCG equals unity. Maintaining VCG at unity can enhance the ZVS range under light loading conditions in CV mode [13]. Therefore, this two-step process ensures the optimal performance of the DAB converter in both CC and CV modes of charging.

In this paper, the operating region of CC mode is split into p points. The optimization problem undertaken in this work is formulated and is presented in (10)-(16).

$$\text{Minimize: } \frac{1}{p} \sum_{i=1}^p I_{(i)lk} \quad (10)$$

$$\text{Subject to: } I_o = I_{o(cc)} \quad (11)$$

$$I_1 > I_{zvs-sec} \quad (12)$$

$$I_2 > I_{zvs-pri} \quad (13)$$

$$d_{min} < d < d_{max} \quad (14)$$

$$\text{Variables: } n \in (n_{min}, n_{max}) \quad (15)$$

$$L_k \in (L_k(min), L_k(max)) \quad (16)$$

Here, (10) is the objective function that calculates the average value of I_{lk} , i.e. $\bar{I}_{lk}$, over the operating range. The constraints are shown in (11)-(14) and the decision variables with their bounds are given in (15) and (16). The optimization algorithm to select the optimal values of the HFT's turns ratio (n_{opt}) and leakage inductance ($L_{k(opt)}$) is depicted in Fig. 4. The input dc-link voltage of the DAB converter to be supplied during CV mode for maintaining VCG at unity is decided by (17)

$$V_{in(CV)} = \left\lceil \frac{V_{o(max)}}{n_{opt}} \right\rceil \quad (17)$$

III. MODE TRANSITION METHOD FOR CC-CV CHARGING

The control block diagram derived from the small-signal model of the DAB converter for closed-loop control in CC-CV charging is shown in Fig. 5; where A_c and B_c are given as [1]

$$A_c = \frac{T_{hs}}{nL_k} (1 - 2|d|) \quad (18)$$

$$B_c = \frac{T_{hs}}{nL_k} d(1 - |d|) \quad (19)$$

The open loop control-to-output transfer functions for CC and CV modes are obtained and given in (20) and (21), respectively.

$$\frac{\hat{i}_o}{\hat{d}} = A_c V_{in} \frac{Z_{out}}{Z_{bat}} \quad (20)$$

$$\frac{\hat{v}_o}{\hat{d}} = A_c V_{in} Z_{out} \quad (21)$$

where,

$$Z_{out} = \frac{Z_{bat}}{sC_o Z_{bat} + 1}$$

The closed-loop proportional-integral (PI) controllers for CC and CV modes are expressed as follows:

$$C_i = K_{p(cc)} + \frac{K_{i(cc)}}{s} \quad (22)$$

$$C_v = K_p(c_v) + \frac{K_i(c_v)}{s} \quad (23)$$

The mode transition with the proposed design of the DAB converter is achieved in two phases. During the first phase, the transition between C_i and C_v controllers is done at the mode transition boundary. In the second phase, the input dc-link voltage is changed to maintain the VCG at unity.

In the steady state of the CC mode of the DAB converter, the error generated by the feedback loop is almost zero. Therefore, the output of the PI controller (d) in this mode will

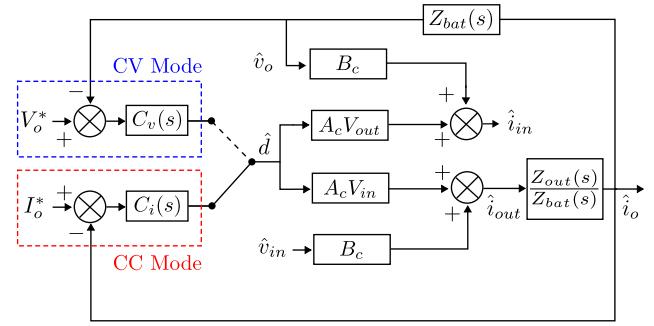


Fig. 5: Control block diagram of the DAB converter.

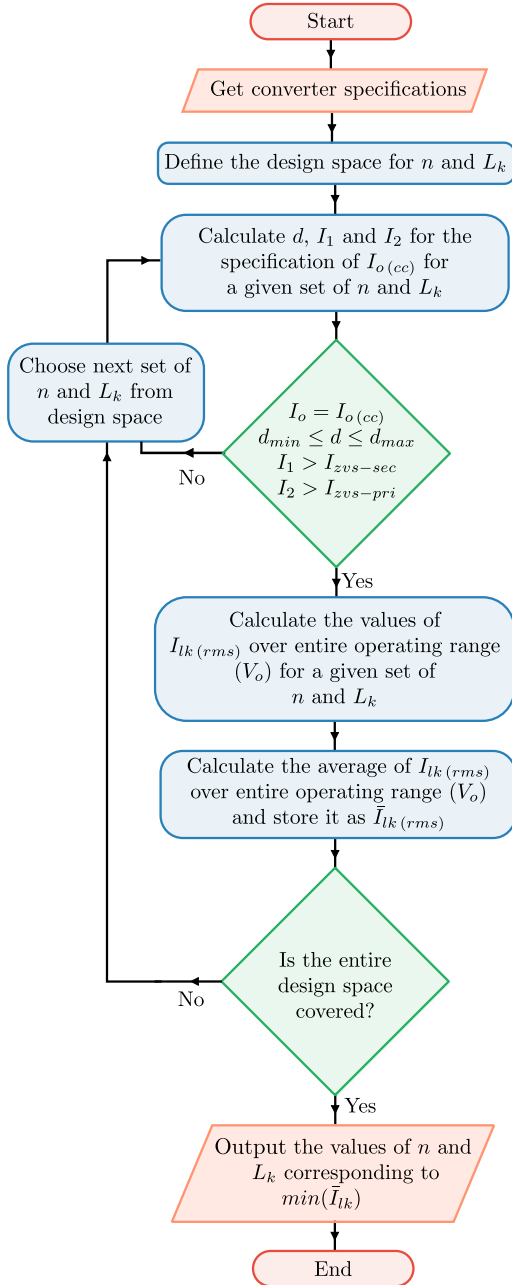


Fig. 4: Flowchart of the proposed loss minimization approach for DAB.

be contributed majorly by the integral term. To attain a smooth mode-transition, the integral terms of C_i and C_v must match. For the second phase of mode-transition, the dc-link voltage must be slowly ramped up to the desired value instead of using a step change to ensure there are no undesirable transients.

IV. RESULTS AND DISCUSSION

For a close comparison, the performance of the DAB converter using the proposed design approach is compared with a few existing design approaches reported in [5], [9]. The values of n and L_k (Fig. 1) obtained using the design technique reported in [5] are found to be equal to 0.1758 and 190 μH and that with the design technique reported in [9] are found to be equal to 0.1732 and 83 μH . With the design approach proposed in this paper, the same are found to be equal to 0.1732 and 83 μH . Based on these results, the values of I_{lk} in CC mode are computed using (4) for the design approaches reported in [5], [9] and this paper and the results are shown in Fig. 6. The value of I_{lk} in CC mode obtained from the design technique proposed in [5] is the highest and that obtained from design technique proposed in [9] and this paper are at par.

With these values obtained, further investigations are carried out in the CV mode of charging. Following (7), the ZVS analysis is carried out on the minimum value of I_{lk} required to realize ZVS for secondary and primary bridges and given in (24) and (25), respectively.

$$I_{1(zvs)} = I_1 - 2V_o \sqrt{\frac{C_{eq-sec}}{L_k}} \quad (24)$$

$$I_{2(zvs)} = I_2 - 2V_{in} \sqrt{\frac{C_{eq-pri}}{L_k}} \quad (25)$$

The results of ZVS analysis obtained from (24) and (25) for secondary and primary bridges are shown in Fig. 7. It can be seen from Fig. 7 that the design approaches [5], [9] do not operate in ZVS region for a wide range of I_o in CV mode. It may be noted here that, though the design approach reported in [9] performs on par in the CC mode with the design approach presented in this paper, it results in inferior performance in the CV mode.

It is already explained in section II that, the dc-link reference can be changed as per (17) to attain unity VCG to ensure ZVS

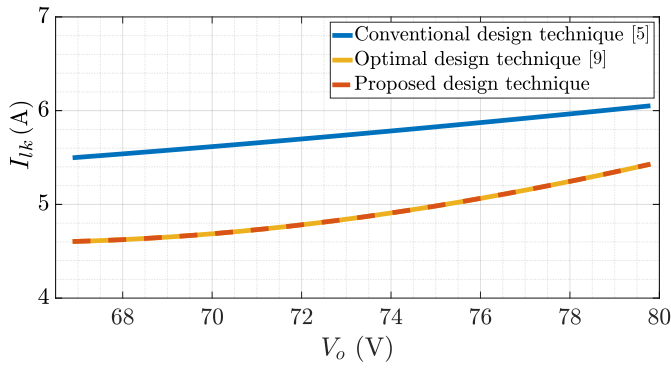


Fig. 6: Comparison of $I_{lk}(rms)$ during CC mode obtained from different design techniques.

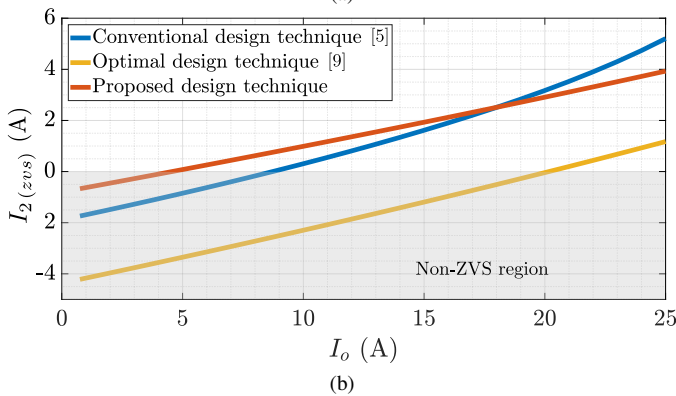
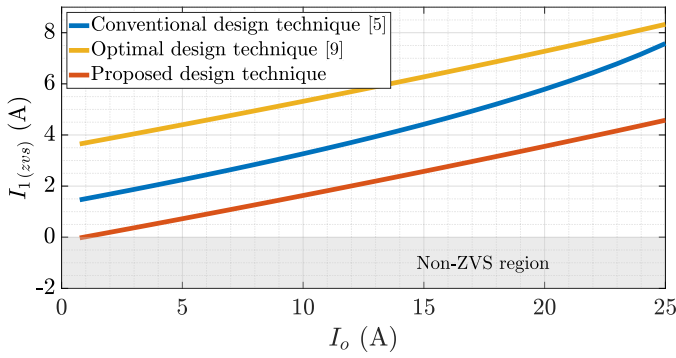
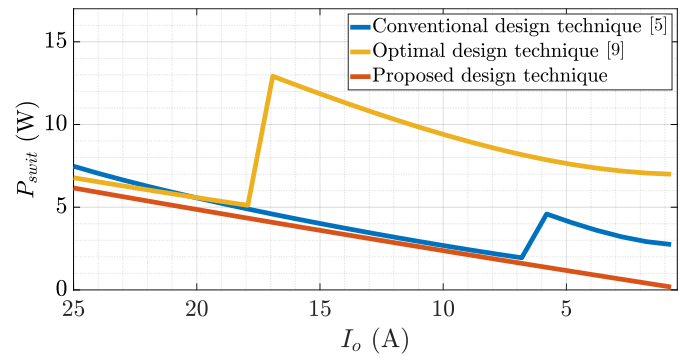


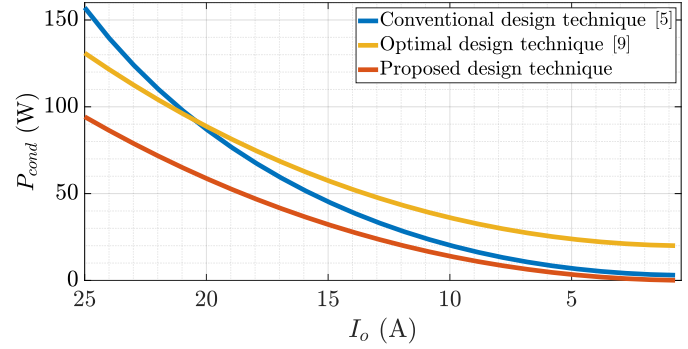
Fig. 7: Comparison of (a) $I_{1(zvs)}$ (b) $I_{2(zvs)}$ during CV mode obtained from different design techniques.

realization and thus minimize switching power losses, The switching power losses (P_{swit}) of the DAB converter using the three design technique discussed above are analytically obtained and are shown in Fig. 8(a) for CV mode. The switching power losses so obtained are least with the proposed design technique compared to the design techniques reported in [5], [9]. Importantly, the conduction power losses of the DAB converter using the design techniques mentioned above are evaluated analytically using (8) and are shown in Fig. 8(b). The results in Fig. 8(b) clearly show that the conduction power losses (P_{cond}) are also reduced by using the proposed design technique over the design techniques reported in [5], [9].

The effectiveness of the proposed design approach on mode



(a) P_{swit} vs I_o curves



(b) P_{cond} vs I_o curves

Fig. 8: Analysis of (a) P_{swit} (b) P_{cond} during CV mode of charging for different design techniques.

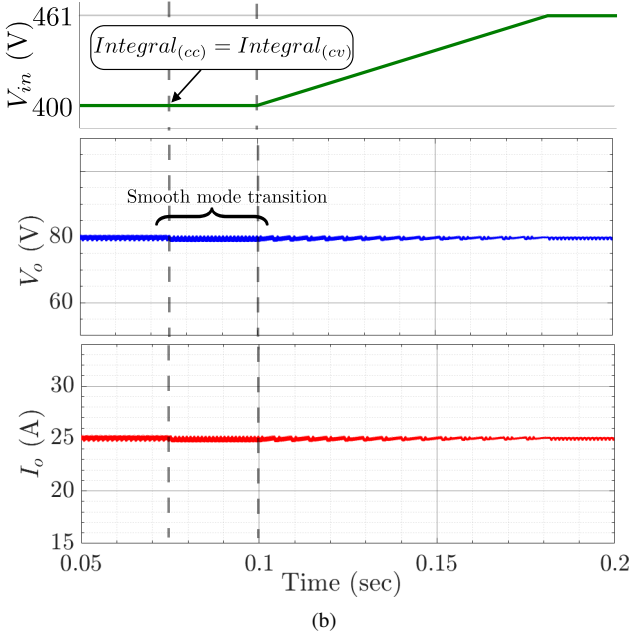
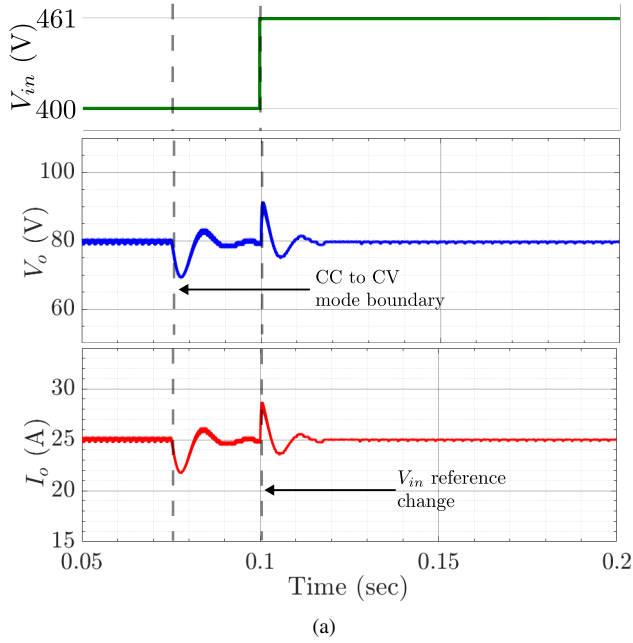
transition compared to the existing methods is also studied in this paper. In this study, the output voltage and output current waveforms of the DAB converter are captured and the results are shown in Fig. 9. It can be seen from Fig. 9 that, during the CC to CV mode transition, undesirable effects of transients are observed in both the V_o and I_o waveforms which would unnecessarily elevate the voltage and current stresses in the switching devices. The scheme showing the traditional approach under mode transition and the approach proposed in this paper are both depicted in the top plots of Fig. 9(a) and Fig. 9(b). This proposed simple approach ensures smooth mode transition as can be seen from results shown in Fig. 9(b).

V. CONCLUSION

This paper introduces a loss minimization approach for the DAB converter over the entire CC-CV modes of operation through design optimization. The proposed design technique results in ZVS enhancement and lower conduction losses of the DAB converter during CV mode while ensuring optimal performance in CC mode in comparison to the existing design approaches. A control-based smooth mode transition process is also presented for the proposed design of the DAB converter. The proposed control-based mode transition method ensures that the undesired transients, which can cause voltage or current stresses in the DAB converter, are completely avoided during CC-CV charging. In this approach, a seamless mode transition is achieved from CC to CV mode without any

TABLE II: Comparison of mode transition methods in literature for CC-CV charging

Reference	Transition Method	Remarks
[11]	Schmitt trigger-based smooth transition circuit	Requirement of additional operational amplifier based comparator circuit
[12]	Transistor-based smooth transition circuit	Requirement of additional transistor-based switching circuits
Proposed	Control-based mode transition	Method is implemented on the DAB converter's controller without an auxiliary circuit

Fig. 9: Comparison of V_o and I_o (a) abrupt mode transition (b) proposed mode transition

additional hardware circuits. The proposed approach can be extended for advanced modulation techniques for DAB converters and different battery charging methods.

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